

Chapter: Greedy & Two Pointers Approach

Core idea: Sort + two pointers + greedy.

Initialization & Setup

Sort arr.

lp → smallest unused item

rp → largest unused item

s → total money spent

level → s / x

bonus → total bonus points

Key Formula

$$\text{need} = (\text{level} + 1) * x - s;$$

This is the minimum price needed to increase loyalty level.

Greedy Logic Implementation

If $\text{arr}[\text{rp}] \geq \text{need}$:

- Largest item can increase loyalty.
- Buy it → earn $\text{arr}[\text{rp}]$ bonus points.
- Move $\text{rp}--$.

Otherwise:

- No remaining item can increase loyalty.
- Buy the smallest item $\text{arr}[\text{lp}]$ to waste minimum money.
- Move $\text{lp}++$.

Complete Implementation (C++)

The full code is preserved exactly as written, with proper indentation and spacing.

```
#include<bits/stdc++.h>
using namespace std;

int main(){

    int t;
    cin>>t;

    while(t--){
        int n,x;
        cin>>n>>x;

        vector<int> arr(n);
        vector<int> ans;

        for(int i=0;i<n;i++){
            cin>>arr[i];
        }

        sort(arr.begin(),arr.end());
        int rp=n-1;
        int lp=0;

        long long bonus=0;
        long long level=0;
        long long s=0;

        while(lp<=rp){
            long long need=((level+1)*x)-s;
            if(arr[rp]>=need){
                s+=arr[rp];
                ans.push_back(arr[rp]);
                bonus+=arr[rp];
                level=s/x;
                rp--;
            }

            else{
                s+=arr[lp];
                ans.push_back(arr[lp]);
                level=s/x;
                lp++;
            }
        }

        cout<<bonus<<"\n";
```